

ECE5424 Final Project

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```
[1]: import os, sys
from pathlib import Path

PROJECT_DIR = '/content/ECE5424Project-main'
if not Path(PROJECT_DIR).exists():
    from google.colab import files
    print('Upload ECE5424Project-main.zip when prompted.')
    uploaded = files.upload()
    [zip_name] = list(uploaded.keys())
    !unzip -q -o "$zip_name" -d /content/

os.chdir(PROJECT_DIR)
sys.path.insert(0, PROJECT_DIR)
print('cwd:', os.getcwd())
!ls
```

```
cwd: /content/ECE5424Project-main
'=4.0'
calibration.py
chat.py
confidence_meter.py
data.py
'ECE5424 Final Project.ipynb'
'ECE5424Project-main (3).zip'
freeform.py
__init__.py
Llama_3_1_8b+_Unsloth_2x_faster_finetuning.ipynb
metrics.py
model.py
pipeline.py
plots.py
README.md
requirements.txt
run_baseline.py
```

```
run_calibration.py
run_chat.py
run_freeform_calibration.py
scripts
src
TruthfulQA.csv
```

```
[2]: !pip install -q -r requirements.txt
!pip install -q --upgrade transformers accelerate
!pip install -q gradio>=4.0
import torch
print('torch', torch.__version__, '| cuda', torch.cuda.is_available(),
      '| device', torch.cuda.get_device_name(0) if torch.cuda.is_available()
      ↪else 'cpu')
```

```
torch 2.10.0+cu128 | cuda True | device NVIDIA RTX PRO 6000 Blackwell Server
Edition
```

```
[3]: from google.colab import userdata
      userdata.get('YourToken') #Use your token here
```

```
-----
SecretNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_9757/2295858123.py in <cell line: 0>()
      1 from google.colab import userdata
----> 2 userdata.get('YourToken') #Use your token here

/usr/local/lib/python3.12/dist-packages/google/colab/userdata.py in get(key)
     66     raise TimeoutException(key)
     67     if not resp.get('exists', False):
--> 68     raise SecretNotFoundError(key)
     69     if not resp.get('access', False):
     70     raise NotebookAccessError(key)

SecretNotFoundError: Secret YourToken does not exist.
```

```
[7]: from google.colab import userdata
      tok = userdata.get('HFToken') #Use your token here
      os.environ['HF_TOKEN'] = tok
      print('Token loaded:', 'yes' if tok else 'no')
```

```
Token loaded: yes
```

```
[8]: MODEL_NAME = 'meta-llama/Meta-Llama-3.1-8B-Instruct'
      BATCH_SIZE = 32
      DTYPE = 'bfloat16'
```

```
!python scripts/run_baseline.py --model {MODEL_NAME} --batch-size {BATCH_SIZE}
  ↪--dtype {DTYPE}
!python scripts/run_calibration.py --tag truthfulqa
```

```
Loading TruthfulQA from /content/ECE5424Project-main/TruthfulQA.csv ...
Built 663 multiple-choice questions (4 options each)
Loading HuggingFace model meta-llama/Meta-Llama-3.1-8B-Instruct (this may take a
while)...
```

config.json:	100%	855/855	[00:00<00:00, 4.58MB/s]
tokenizer_config.json:		55.4kB	[00:00, 31.9MB/s]
tokenizer.json:		9.09MB	[00:00, 19.0MB/s]
special_tokens_map.json:	100%	296/296	[00:00<00:00, 1.60MB/s]

```
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
model.safetensors.index.json: 23.9kB [00:00, 39.1MB/s]
Downloading (incomplete total...): 0.00B [00:00, ?B/s]
Downloading (incomplete total...): 93% 14.9G/16.1G [00:40<00:02, 532MB/s]
Downloading (incomplete total...): 100% 16.1G/16.1G [00:40<00:00, 991MB/s]
Fetching 4 files: 100% 4/4 [00:40<00:00, 10.15s/it]
Download complete: 100% 16.1G/16.1G [00:40<00:00, 991MB/s]
Loading weights: 100% 291/291 [00:00<00:00, 15537.23it/s]
```

```
generation_config.json: 100% 184/184 [00:00<00:00, 2.37MB/s]
Model loaded on device=cuda, dtype=torch.bfloat16, batch_size=32
```

```
scoring: 0% 0/663 [00:00<?, ?q/s]
scoring: 5% 32/663 [00:05<01:55, 5.47q/s]
scoring: 10% 64/663 [00:05<00:46, 12.83q/s]
scoring: 14% 96/663 [00:06<00:25, 22.27q/s]
scoring: 19% 128/663 [00:06<00:16, 33.13q/s]
scoring: 24% 160/663 [00:06<00:10, 47.43q/s]
Download complete: 100% 16.1G/16.1G [00:50<00:00, 991MB/s]
scoring: 34% 224/663 [00:06<00:05, 79.46q/s]
scoring: 39% 256/663 [00:07<00:04, 98.47q/s]
scoring: 43% 288/663 [00:07<00:03, 120.69q/s]
scoring: 48% 320/663 [00:07<00:02, 142.54q/s]
scoring: 53% 352/663 [00:07<00:01, 156.51q/s]
scoring: 58% 384/663 [00:07<00:01, 178.09q/s]
scoring: 63% 416/663 [00:07<00:01, 184.61q/s]
scoring: 68% 448/663 [00:08<00:01, 173.57q/s]
scoring: 72% 480/663 [00:08<00:00, 189.15q/s]
scoring: 77% 512/663 [00:08<00:00, 202.05q/s]
scoring: 82% 544/663 [00:08<00:00, 201.53q/s]
scoring: 87% 576/663 [00:08<00:00, 192.57q/s]
scoring: 92% 608/663 [00:08<00:00, 186.39q/s]
scoring: 97% 640/663 [00:08<00:00, 189.49q/s]
scoring: 100% 663/663 [00:09<00:00, 72.47q/s]
```

Baseline accuracy: 0.575
Average confidence: 0.773
Overconfidence gap: +0.198
Saved per-question predictions to /content/ECE5424Project-main/results/baseline_predictions.csv
Saved raw logits to /content/ECE5424Project-main/results/baseline_logits.npz
Download complete: 100% 16.1G/16.1G [00:53<00:00, 300MB/s]
Loaded 663 predictions from /content/ECE5424Project-main/results/baseline_logits.npz
Calibration split: 331 val / 332 test

=== Calibration report ===

```
{
  "tag": "truthfulqa",
  "temperature": 2.899286051213417,
  "raw": {
    "n": 332,
    "accuracy": 0.5662650602409639,
    "avg_confidence": 0.7784826371087483,
    "overconfidence_gap": 0.21221757686778442,
    "ece": 0.21221757686778459,
    "brier": 0.26207404674050455
  },
  "temperature_scaled": {
    "n": 332,
    "accuracy": 0.5662650602409639,
    "avg_confidence": 0.5356485028347105,
    "overconfidence_gap": -0.030616557406253353,
    "ece": 0.05362301694976227,
    "brier": 0.21287984106711313
  },
  "temperature_then_isotonic": {
    "n": 332,
    "accuracy": 0.5662650602409639,
    "avg_confidence": 0.5942759935604202,
    "overconfidence_gap": 0.028010933319456255,
    "ece": 0.049757436973738974,
    "brier": 0.21403825769006687
  }
}
```

Artefacts written to /content/ECE5424Project-main/results

```
[9]: import json, pathlib
      print(json.dumps(json.loads(pathlib.Path('results/truthfulqa_report.json').
      ↪read_text()), indent=2))
```

```

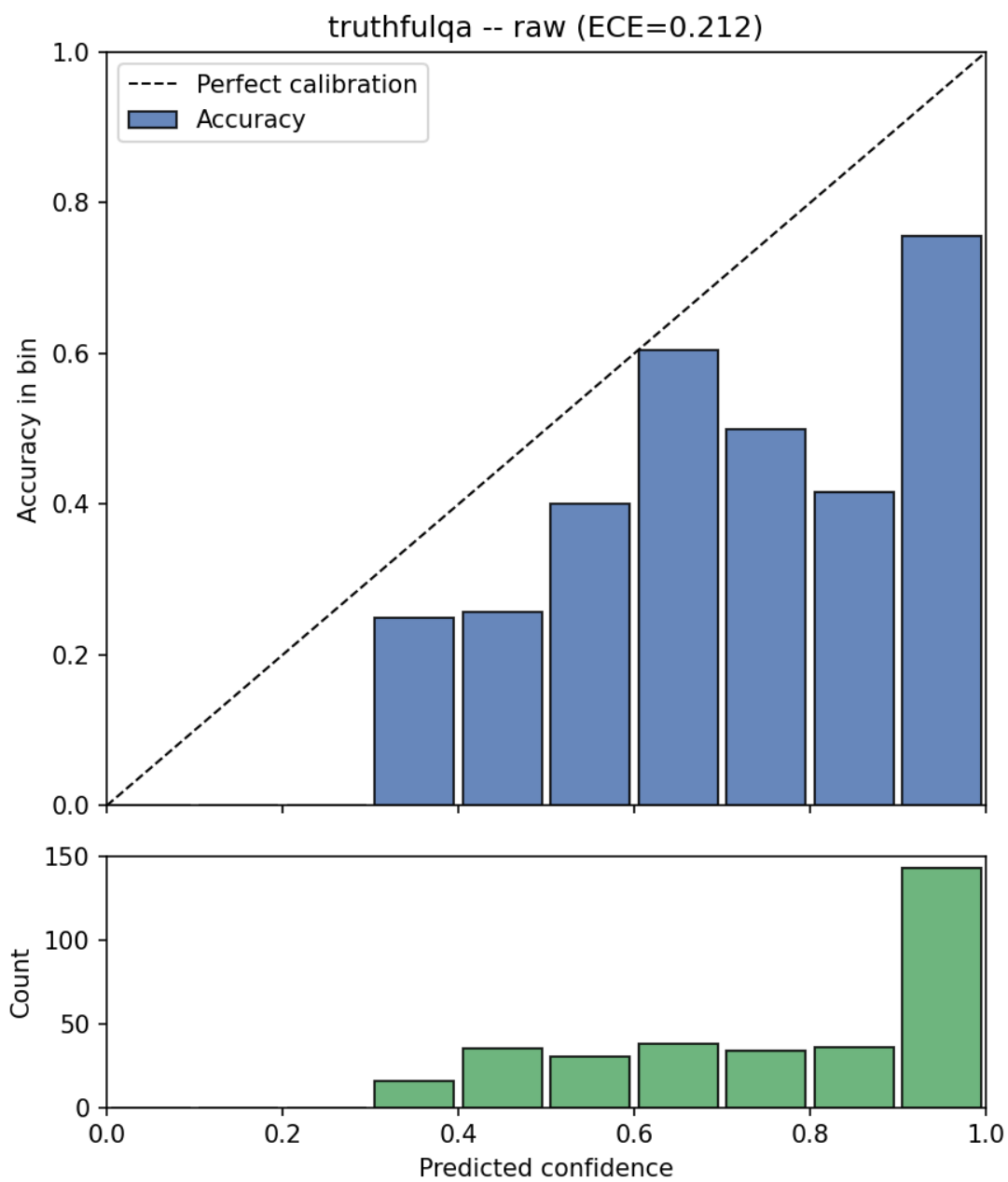
from IPython.display import Image, display
for kind in ('raw', 'temperature', 'isotonic'):
    display(Image(f'results/truthfulqa_reliability_{kind}.png'))

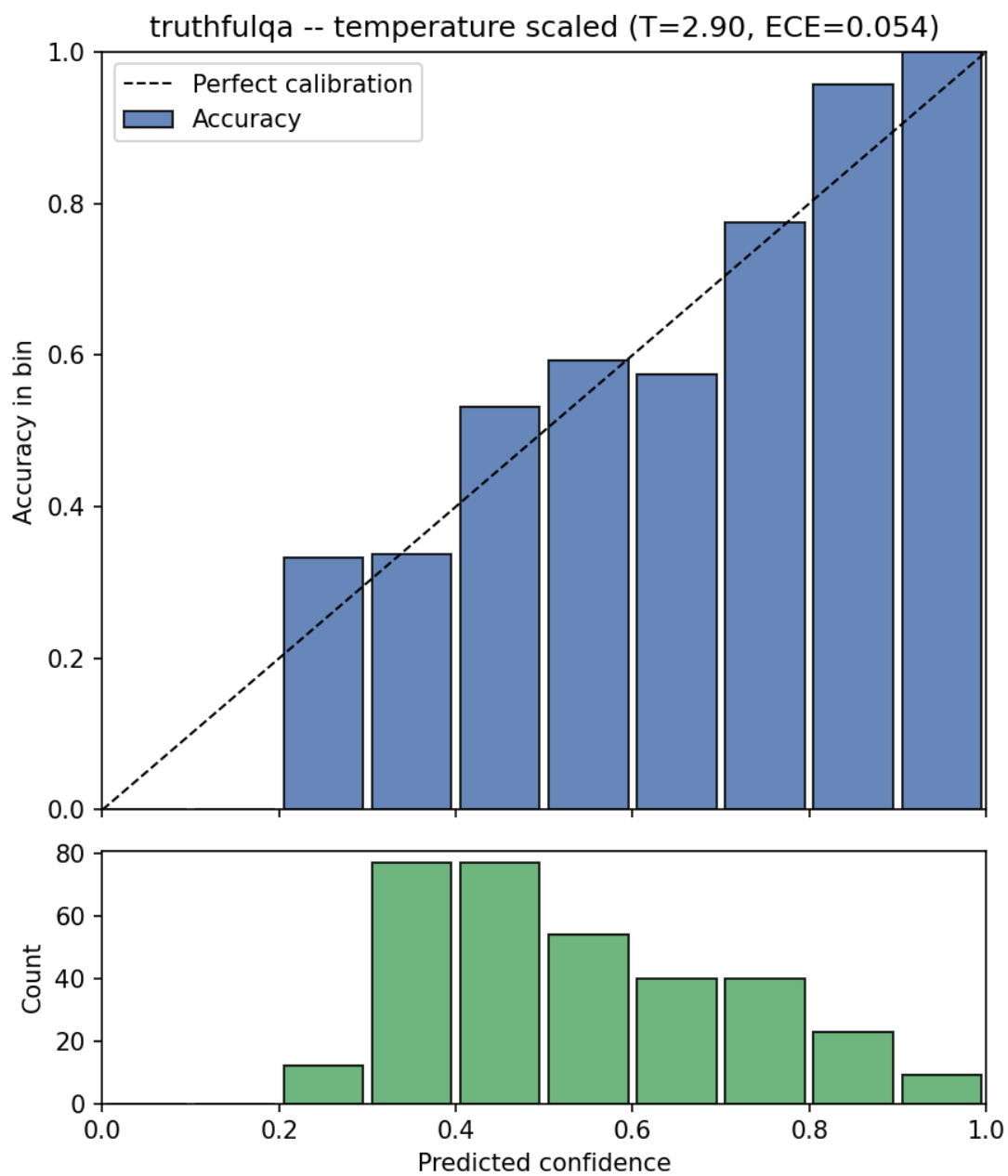
```

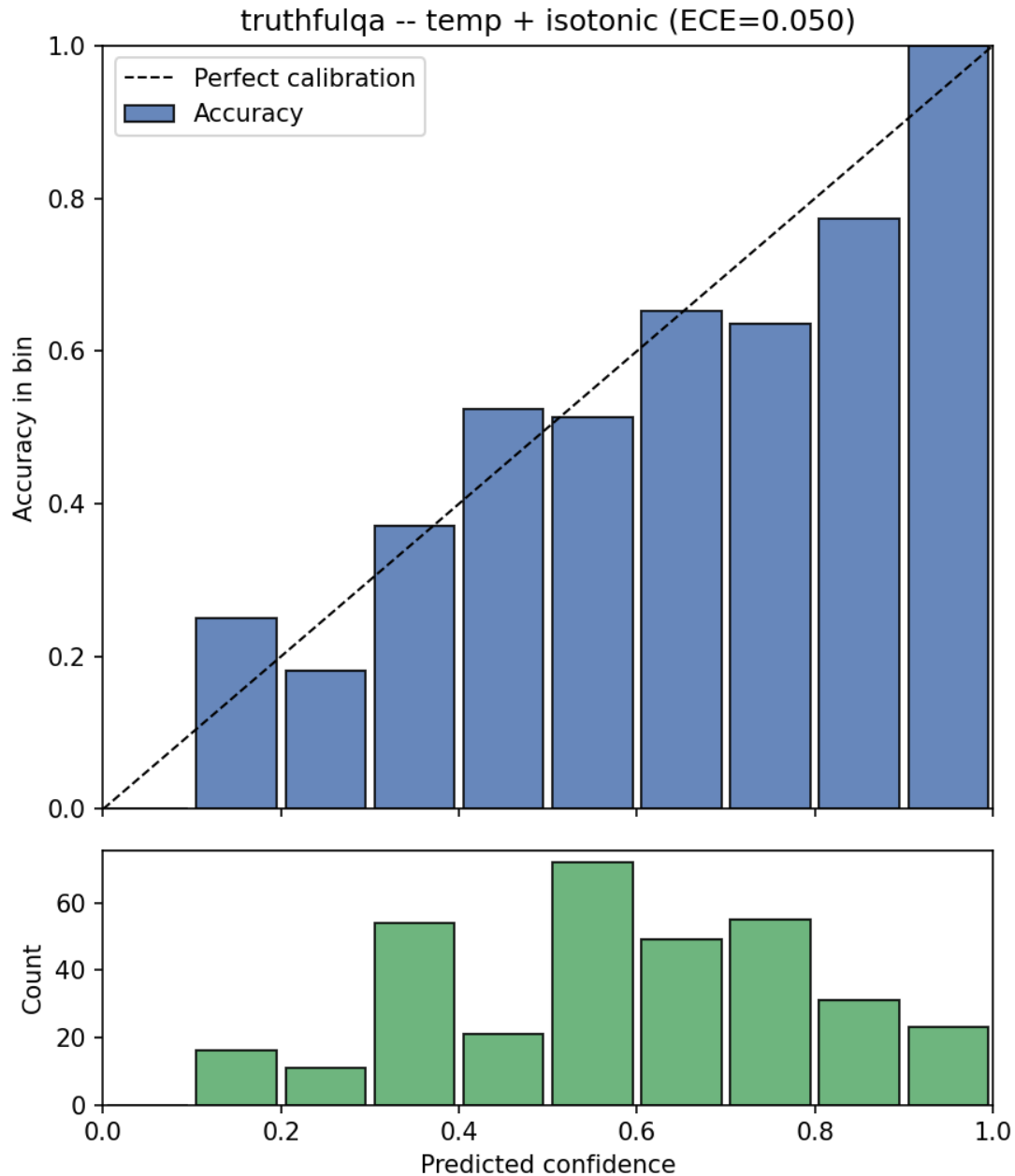
```

{
  "tag": "truthfulqa",
  "temperature": 2.899286051213417,
  "raw": {
    "n": 332,
    "accuracy": 0.5662650602409639,
    "avg_confidence": 0.7784826371087483,
    "overconfidence_gap": 0.21221757686778442,
    "ece": 0.21221757686778459,
    "brier": 0.26207404674050455
  },
  "temperature_scaled": {
    "n": 332,
    "accuracy": 0.5662650602409639,
    "avg_confidence": 0.5356485028347105,
    "overconfidence_gap": -0.030616557406253353,
    "ece": 0.05362301694976227,
    "brier": 0.21287984106711313
  },
  "temperature_then_isotonic": {
    "n": 332,
    "accuracy": 0.5662650602409639,
    "avg_confidence": 0.5942759935604202,
    "overconfidence_gap": 0.028010933319456255,
    "ece": 0.049757436973738974,
    "brier": 0.21403825769006687
  }
}

```







```
[10]: os.environ['HF_TOKEN'] = tok
!python scripts/run_freeform_calibration.py \
--model {MODEL_NAME} --dtype {DTYPE} --max-new-tokens 96
```

Loading TruthfulQA from /content/ECE5424Project-main/TruthfulQA.csv
Using 790 questions
Loading model meta-llama/Meta-Llama-3.1-8B-Instruct on cuda (torch.bfloat16)
[transformers] `torch\_dtype` is deprecated! Use `dtype` instead!


```

Loading weights: 100% 291/291 [00:00<00:00, 14985.17it/s]
generating: 0% 0/790 [00:00<?, ?it/s][transformers] Ignoring
clean_up_tokenization_spaces=True for BPE tokenizer TokenizersBackend. The
clean_up_tokenization post-processing step is designed for WordPiece tokenizers
and is destructive for BPE (it strips spaces before punctuation). Set
clean_up_tokenization_spaces=False to suppress this warning, or set
clean_up_tokenization_spaces_for_bpe_even_though_it_will_corrupt_output=True to
force cleanup anyway.
generating: 100% 790/790 [13:26<00:00, 1.02s/it]
Saved 790 free-form predictions -> /content/ECE5424Project-
main/results/freeform_predictions.csv
Judge coverage: 787/790 (99.6%)
Saved calibrator -> /content/ECE5424Project-
main/results/freeform_calibrator.joblib
Saved report -> /content/ECE5424Project-
main/results/freeform_calibrator_report.json

```

```

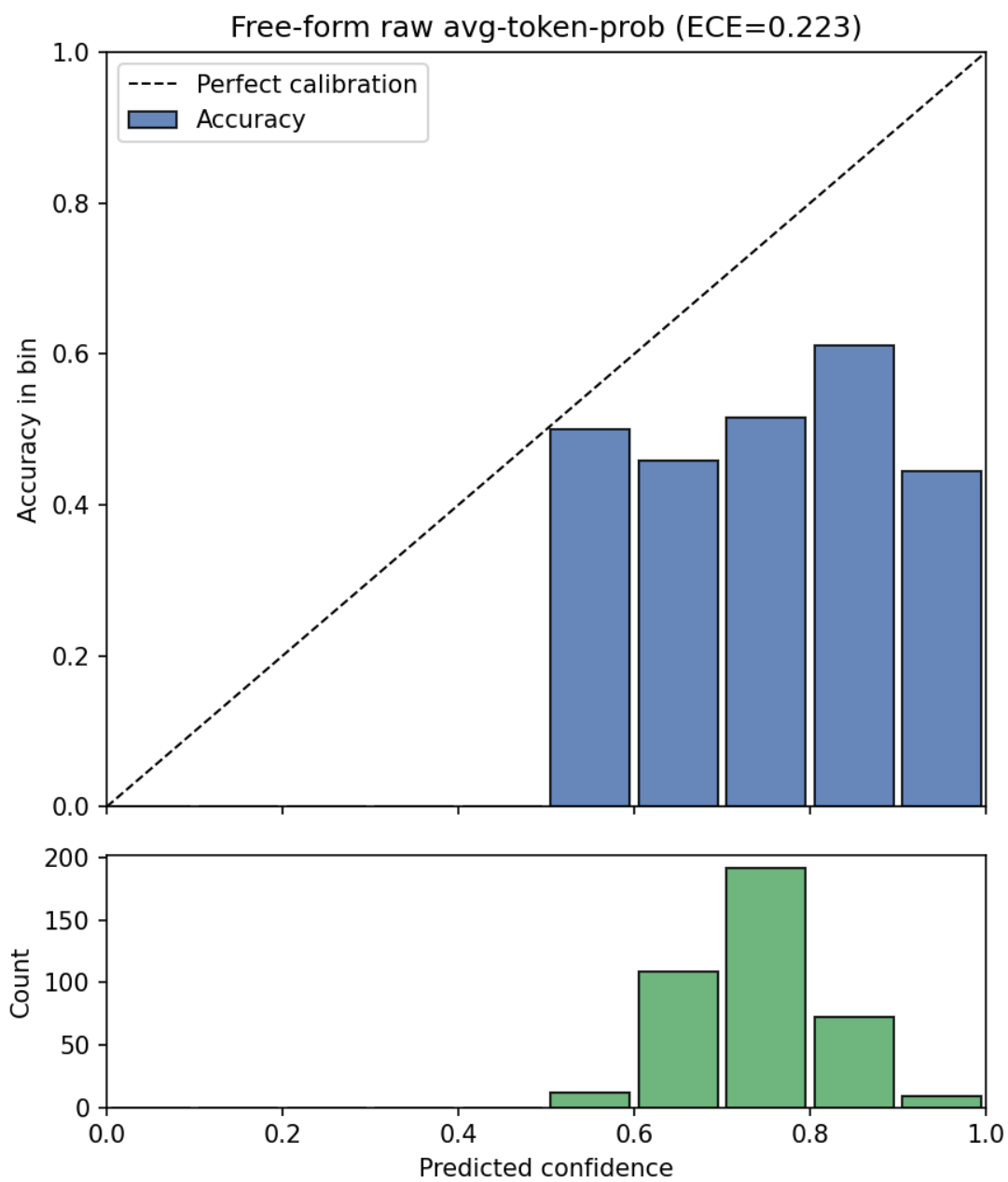
=== Free-form calibration report ===
{
  "model": "meta-llama/Meta-Llama-3.1-8B-Instruct",
  "n_generated": 790,
  "n_judged": 787,
  "raw": {
    "n": 394,
    "accuracy": 0.5152284263959391,
    "avg_confidence": 0.738420838827067,
    "overconfidence_gap": 0.22319241243112786,
    "ece": 0.22319241243112803,
    "brier": 0.2988604024157758
  },
  "calibrated": {
    "n": 394,
    "accuracy": 0.5152284263959391,
    "avg_confidence": 0.4978286113594437,
    "overconfidence_gap": -0.01739981503649546,
    "ece": 0.060296809276426104,
    "brier": 0.2557588014503242
  }
}

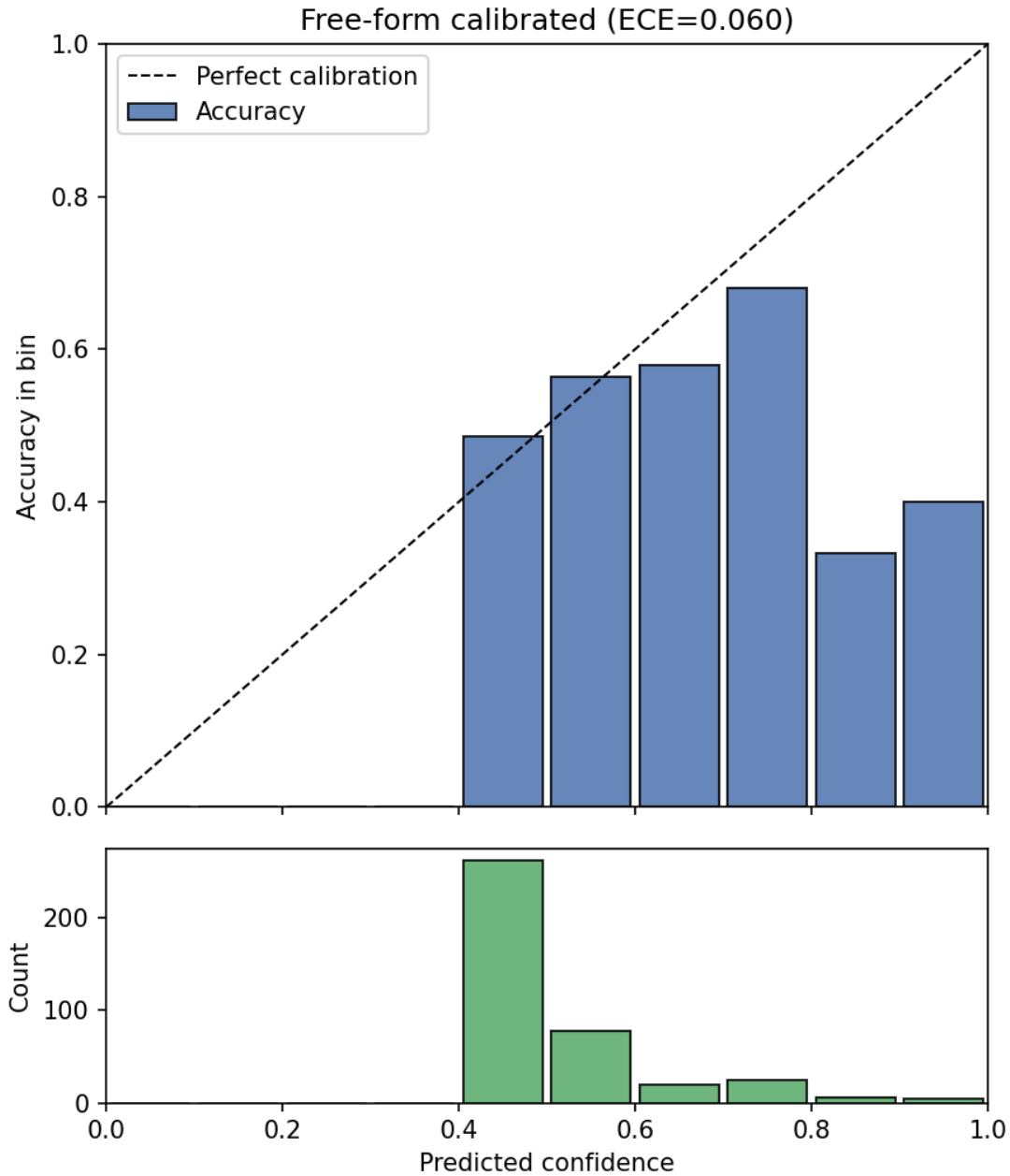
```

```

[14]: for kind in ('raw', 'calibrated'):
      display(Image(f'results/freeform_reliability_{kind}.png'))

```





```
[12]: os.environ['HF_TOKEN'] = tok
      !python scripts/run_chat.py --model {MODEL_NAME} --dtype {DTYPE} --share
```

```
Loading chatbot (model=meta-llama/Meta-Llama-3.1-8B-Instruct)...
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100% 291/291 [00:00<00:00, 15258.50it/s]
Model loaded on cuda. Calibrator: loaded.
* Running on local URL: http://0.0.0.0:7860
```

* Running on public URL: <https://15dd6dcd025dbe3b9d.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run ``gradio deploy`` from the terminal in the working directory to deploy to Hugging Face Spaces (<https://huggingface.co/spaces>)

Keyboard interruption in main thread... closing server.

Traceback (most recent call last):

File `"/usr/local/lib/python3.12/dist-packages/gradio/blocks.py"`, line 3043, in `block_thread`

`time.sleep(0.1)`

KeyboardInterrupt

During handling of the above exception, another exception occurred:

Traceback (most recent call last):

File `"/content/ECE5424Project-main/scripts/run_chat.py"`, line 133, in `<module>`
`main()`

File `"/content/ECE5424Project-main/scripts/run_chat.py"`, line 129, in `main`
`run_gradio(bot, args)`

File `"/content/ECE5424Project-main/scripts/run_chat.py"`, line 101, in `run_gradio`

`demo.queue().launch()`

File `"/usr/local/lib/python3.12/dist-packages/gradio/blocks.py"`, line 2950, in `launch`

`self.block_thread()`

File `"/usr/local/lib/python3.12/dist-packages/gradio/blocks.py"`, line 3045, in `block_thread`

`print("Keyboard interruption in main thread... closing server.")`

KeyboardInterrupt

Killing tunnel 0.0.0.0:7860 <> <https://15dd6dcd025dbe3b9d.gradio.live>

```
[13]: !cd results && zip -qr /content/ece5424_results.zip . && ls -lh /content/
      ↪ ece5424_results.zip
      # from google.colab import files; files.download('/content/ece5424_results.zip')
```

```
-rw-r--r-- 1 root root 558K May  4 00:32 /content/ece5424_results.zip
```

```
[15]: !apt-get update
      !apt-get install -y --fix-missing texlive-xetex
      ↪ texlive-fonts-recommended texlive-plain-generic
```

```
0% [Working]                               Get:1 https://cli.github.com/packages stable
InRelease [3,917 B]
```

```
0% [Connecting to archive.ubuntu.com (91.189.91.83)] [Connecting to security.ub
```

```
0% [Connecting to archive.ubuntu.com (91.189.91.83)] [Connecting to security.ub
```

```
Get:2 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease
```

```
[3,632 B]
```

```
0% [Connecting to archive.ubuntu.com (91.189.91.83)] [Connecting to security.ub
```

Get:3 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64
 InRelease [1,581 B]
 0% [Connecting to archive.ubuntu.com (91.189.91.83)] [Connecting to security.ub
 0% [Connecting to archive.ubuntu.com (91.189.91.83)] [Connecting to security.ub
 0% [Waiting for headers] [Waiting for headers] [Connecting to r2u.stat.illinois
 Get:4 <https://cli.github.com/packages/stable/main/amd64> Packages [356 B]
 Get:5 <https://cloud.r-project.org/bin/linux/ubuntu/jammy-cran40/> Packages [89.0
 kB]
 Hit:6 <http://archive.ubuntu.com/ubuntu/jammy> InRelease
 Get:7 <http://security.ubuntu.com/ubuntu/jammy-security> InRelease [129 kB]
 Get:8 <http://archive.ubuntu.com/ubuntu/jammy-updates> InRelease [128 kB]
 Get:9 <https://r2u.stat.illinois.edu/ubuntu/jammy> InRelease [6,555 B]
 Get:10 <https://r2u.stat.illinois.edu/ubuntu/jammy/main/amd64> Packages [2,991 kB]
 Get:11 <http://archive.ubuntu.com/ubuntu/jammy-backports> InRelease [127 kB]
 Get:12 <http://security.ubuntu.com/ubuntu/jammy-security/universe/amd64> Packages
 [1,292 kB]
 Get:13 <http://archive.ubuntu.com/ubuntu/jammy-updates/universe/amd64> Packages
 [1,601 kB]
 Get:14 <http://security.ubuntu.com/ubuntu/jammy-security/main/amd64> Packages
 [3,864 kB]
 Get:15 <http://archive.ubuntu.com/ubuntu/jammy-updates/multiverse/amd64> Packages
 [86.0 kB]
 Get:16 <http://archive.ubuntu.com/ubuntu/jammy-updates/restricted/amd64> Packages
 [7,225 kB]
 Get:17 <http://security.ubuntu.com/ubuntu/jammy-security/multiverse/amd64>
 Packages [61.6 kB]
 Get:18 <http://security.ubuntu.com/ubuntu/jammy-security/restricted/amd64>
 Packages [6,862 kB]
 Get:19 <http://archive.ubuntu.com/ubuntu/jammy-updates/main/amd64> Packages [4,237
 kB]
 Get:20 <https://r2u.stat.illinois.edu/ubuntu/jammy/main/all> Packages [10.1 MB]
 Get:21 <http://archive.ubuntu.com/ubuntu/jammy-backports/universe/amd64> Packages
 [39.0 kB]
 Get:22 <http://archive.ubuntu.com/ubuntu/jammy-backports/main/amd64> Packages [112
 kB]
 Get:23
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64
 Packages [2,608 kB]
 Ign:24 <https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu/jammy> InRelease
 Ign:25 <https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu/jammy>
 InRelease
 Ign:26 <https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu/jammy> InRelease
 Ign:24 <https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu/jammy> InRelease
 Ign:25 <https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu/jammy>
 InRelease
 Ign:26 <https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu/jammy> InRelease
 Ign:24 <https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu/jammy> InRelease
 Ign:25 <https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu/jammy>

```

InRelease
Ign:26 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
Err:24 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
  Could not connect to ppa.launchpadcontent.net:443 (185.125.190.80), connection
  timed out
Err:25 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy
InRelease
  Unable to connect to ppa.launchpadcontent.net:443:
Err:26 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
  Unable to connect to ppa.launchpadcontent.net:443:
Fetched 41.6 MB in 37s (1,115 kB/s)
Reading package lists... Done
W: Skipping acquire of configured file 'main/source/Sources' as repository
  'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide
  it (sources.list entry misspelt?)
W: Failed to fetch
https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu/dists/jammy/InRelease
Could not connect to ppa.launchpadcontent.net:443 (185.125.190.80), connection
  timed out
W: Failed to fetch https://ppa.launchpadcontent.net/graphics-
  drivers/ppa/ubuntu/dists/jammy/InRelease Unable to connect to
  ppa.launchpadcontent.net:443:
W: Failed to fetch
https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu/dists/jammy/InRelease
Unable to connect to ppa.launchpadcontent.net:443:
W: Some index files failed to download. They have been ignored, or old ones used
  instead.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package texlive-fonts-recommended_┐texlive-plain-generic

```

```

[16]: !apt-get -qq install texlive texlive-xetex texlive-latex-extra pandoc
      !pip install --quiet pypandoc

```

```

Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 122354 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...

```

```

Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...

```

```

Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.12_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.12) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.12_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.12) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.3_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../32-libtexluajit2_2021.20210626.59705-1ubuntu0.3_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package libzip-0-13:amd64.

```



```

Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.3_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.3) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...

```

```

Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...

```

```

Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.3) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.3) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.12) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.12) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

```

```

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

```

```

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.

```

```

[18]: [!]jupyter nbconvert --to pdf "/content/drive/My Drive/ECE5424Project-main/
↪notebooks/ECE5424 Final Project.ipynb"

```

```

[NbConvertApp] WARNING | pattern '/content/drive/My Drive/ECE5424Project-
main/notebooks/ECE5424 Final Project.ipynb' matched no files
This application is used to convert notebook files (*.ipynb)
to various other formats.

```

```

WARNING: THE COMMANDLINE INTERFACE MAY CHANGE IN FUTURE RELEASES.

```

Options

=====

The options below are convenience aliases to configurable class-options,

as listed in the "Equivalent to" description-line of the aliases.

To see all configurable class-options for some <cmd>, use:

```
<cmd> --help-all
```

--debug

set log level to logging.DEBUG (maximize logging output)

Equivalent to: [--Application.log_level=10]

--show-config

Show the application's configuration (human-readable format)

Equivalent to: [--Application.show_config=True]

--show-config-json

Show the application's configuration (json format)

Equivalent to: [--Application.show_config_json=True]

--generate-config

generate default config file

Equivalent to: [--JupyterApp.generate_config=True]

-y

Answer yes to any questions instead of prompting.

Equivalent to: [--JupyterApp.answer_yes=True]

--execute

Execute the notebook prior to export.

Equivalent to: [--ExecutePreprocessor.enabled=True]

--allow-errors

Continue notebook execution even if one of the cells throws an error and include the error message in the cell output (the default behaviour is to abort conversion). This flag is only relevant if '--execute' was specified, too.

Equivalent to: [--ExecutePreprocessor.allow_errors=True]

--stdin

read a single notebook file from stdin. Write the resulting notebook with default basename 'notebook.*'

Equivalent to: [--NbConvertApp.from_stdin=True]

--stdout

Write notebook output to stdout instead of files.

Equivalent to: [--NbConvertApp.writer_class=StdoutWriter]

--inplace

Run nbconvert in place, overwriting the existing notebook (only relevant when converting to notebook format)

Equivalent to: [--NbConvertApp.use_output_suffix=False

--NbConvertApp.export_format=notebook --FilesWriter.build_directory=]

--clear-output

Clear output of current file and save in place, overwriting the existing notebook.

Equivalent to: [--NbConvertApp.use_output_suffix=False

--NbConvertApp.export_format=notebook --FilesWriter.build_directory=

--ClearOutputPreprocessor.enabled=True]

--coalesce-streams

Coalesce consecutive stdout and stderr outputs into one stream (within each cell).

Equivalent to: [--NbConvertApp.use_output_suffix=False
--NbConvertApp.export_format=notebook --FilesWriter.build_directory=
--CoalesceStreamsPreprocessor.enabled=True]

--no-prompt
Exclude input and output prompts from converted document.
Equivalent to: [--TemplateExporter.exclude_input_prompt=True
--TemplateExporter.exclude_output_prompt=True]

--no-input
Exclude input cells and output prompts from converted document.
This mode is ideal for generating code-free reports.
Equivalent to: [--TemplateExporter.exclude_output_prompt=True
--TemplateExporter.exclude_input=True
--TemplateExporter.exclude_input_prompt=True]

--allow-chromium-download
Whether to allow downloading chromium if no suitable version is found on the system.
Equivalent to: [--WebPDFExporter.allow_chromium_download=True]

--disable-chromium-sandbox
Disable chromium security sandbox when converting to PDF..
Equivalent to: [--WebPDFExporter.disable_sandbox=True]

--show-input
Shows code input. This flag is only useful for dejavu users.
Equivalent to: [--TemplateExporter.exclude_input=False]

--embed-images
Embed the images as base64 dataurls in the output. This flag is only useful for the HTML/WebPDF/Slides exports.
Equivalent to: [--HTMLExporter.embed_images=True]

--sanitize-html
Whether the HTML in Markdown cells and cell outputs should be sanitized..
Equivalent to: [--HTMLExporter.sanitize_html=True]

--log-level=<Enum>
Set the log level by value or name.
Choices: any of [0, 10, 20, 30, 40, 50, 'DEBUG', 'INFO', 'WARN', 'ERROR', 'CRITICAL']
Default: 30
Equivalent to: [--Application.log_level]

--config=<Unicode>
Full path of a config file.
Default: ''
Equivalent to: [--JupyterApp.config_file]

--to=<Unicode>
The export format to be used, either one of the built-in formats
['asciidoc', 'custom', 'html', 'latex', 'markdown', 'notebook', 'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides', 'webpdf']
or a dotted object name that represents the import path for an
``Exporter`` class
Default: ''
Equivalent to: [--NbConvertApp.export_format]

copy of reveal.js: e.g., "reveal.js".
 If a relative path is given, it must be a subdirectory of the
 current directory (from which the server is run).
 See the usage documentation
 ([https://nbconvert.readthedocs.io/en/latest/usage.html#reveal-js-
 html-slideshow](https://nbconvert.readthedocs.io/en/latest/usage.html#reveal-js-html-slideshow))
 for more details.
 Default: ''
 Equivalent to: [--SlidesExporter.reveal_url_prefix]
 --nbformat=<Enum>
 The nbformat version to write.
 Use this to downgrade notebooks.
 Choices: any of [1, 2, 3, 4]
 Default: 4
 Equivalent to: [--NotebookExporter.nbformat_version]

Examples

The simplest way to use nbconvert is

```
> jupyter nbconvert mynotebook.ipynb --to html
```

Options include ['asciidoc', 'custom', 'html', 'latex', 'markdown',
 'notebook', 'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides',
 'webpdf'].

```
> jupyter nbconvert --to latex mynotebook.ipynb
```

Both HTML and LaTeX support multiple output templates. LaTeX
 includes

'base', 'article' and 'report'. HTML includes 'basic', 'lab' and
 'classic'. You can specify the flavor of the format used.

```
> jupyter nbconvert --to html --template lab mynotebook.ipynb
```

You can also pipe the output to stdout, rather than a file

```
> jupyter nbconvert mynotebook.ipynb --stdout
```

PDF is generated via latex

```
> jupyter nbconvert mynotebook.ipynb --to pdf
```

You can get (and serve) a Reveal.js-powered slideshow

```
> jupyter nbconvert myslides.ipynb --to slides --post serve
```


Multiple notebooks can be given at the command line in a couple of different ways:

```
> jupyter nbconvert notebook*.ipynb
> jupyter nbconvert notebook1.ipynb notebook2.ipynb
```

or you can specify the notebooks list in a config file, containing::

```
c.NbConvertApp.notebooks = ["my_notebook.ipynb"]
```

```
> jupyter nbconvert --config mycfg.py
```

To see all available configurables, use `--help-all`.

```
[ ]: from google.colab import drive
drive.mount('/content/drive')
```